Joy Zhang

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EDUCATION

Cornell University, College of Engineering
Master of Engineering, Computer Science, GPA 3.9 **May 2021**
Bachelor of Science, Computer Science Major, Game Design Minor, GPA 3.5 **Dec 2020**

PROFESSIONAL EXPERIENCE

Sandtable **May 2025 – April 2026**
Game Engineer

- Developed a cross-platform (PC, tablet, web, mobile) multiplayer strategy game in **Unity** by designing game systems and implementing the core gameplay loop, game mechanics, and networked gameplay
- Created a versatile and easily-customizable UI and theme system to streamline UI development
- Wrote shaders and added in-game visuals and effects such as blur to the render pipeline
- Used real-world data and maps to set the game anywhere in the world, performed complex geospatial calculations, and explored standard geospatial libraries such as PROJ, GDAL, GeoJSON, and EPSG

Hometopia Inc.
Senior Game Developer **Dec 2024 - Feb 2025**
Junior Game Developer **Jul 2024 - Dec 2024**

- Contributed to the development of an online multiplayer house-building PC sandbox game in **Unity** by writing clean, well-organized, and easily maintainable **C#** code
- Took ownership of UI development, collaborated with designers to give game a cohesive look and feel
- Lead and mentored a small team of junior developers, delegated tasks, and performed code reviews
- Developed an audio system to centralize control of music, ambience, and sound effects
- Created a system for the first-time user experience and tutorial
- Improved multiplayer features, including friend list, parties, chat, and save file transfer and sharing
- Investigated and fixed critical gameplay, multiplayer, and UI bugs, and developed new features

Northrop Grumman
ODIN, *Game Developer* **Jan 2022 - Jun 2024**

- Developed Omniverse apps in **Python** with Universal Scene Description to build a virtual warfighting environment by controlling drones and sensors, streamlining data generation, and visualizing data
- Built physically accurate flight simulators for Volansi and Spirit drones with Omniverse's integrated **PhysX** physics engine and performed manual 3D vector math calculations
- Integrated Omniverse simulation with external AI code and live drones running in real-time
- Created a data generation pipeline for Omniverse to train AI/ML models by developing a suite of tools in **Python** to create, control, and randomize scenes, assets, and perceptual variables

COAGEN, *Software Engineer* **Aug 2021 - Dec 2021**

- Built demo in **Java** for a long short-term memory (LSTM) predicting adhoc interruptions to schedule

PROJECTS

Game Design Initiative at Cornell **Jan 2019 – May 2021**

- Developed gameplay mechanics, physics, and programmatic animations with the Cornell University Games Library in **C++** and **libGDX** in **Java** as a gameplay programmer to make games in teams of 7
- Accepted to BostonFIG Fest with computer game *Starstruck* (2019) and mobile game *Spectacle!* (2020)

Global Summit **Jun 2020 – Aug 2020**

- Created **Pygame** simulation for an unmanned wildfire identification and suppression system

Cornell Big Red Hacks **Sep 2019**

- Awarded Best Design Hack for *One Call Away*, a short story platformer created in **Unity** in 36 hours

SKILLS

Competencies: Agile, Algorithms, Data structures, Game development, Gameplay programming, Object-oriented programming, Physics simulation, UI programming
Programming Languages: C, C#, C++, Java, MATLAB, OCaml, Perl, PHP, Python
Specialties: Blender, Box2D, Construct3, libGDX, Nvidia Omniverse, PhysX, Pygame, Unity
Other Software: CSS, Docker, Figma, Git, HTML, Linux, Maven, Robot Framework, Universal Scene Description